

SLIP – 14

Section I: Data Structures [15 Marks]

Q 1) Read the data from “Student.txt (Stud_name, Age, Percentage)” file and sort on Percentage using selection sort.(Create a separate sorted file). Assumed file format of Student.txt (one record per line): Stud_name Age Percentage

Answer:

```
#include <stdio.h>

struct Stud
{
    char name[50];
    int age;    float
per;
};

void selectionSort(struct Stud s[], int n)
{
    int i, j, min;
    struct Stud temp;

    for(i = 0; i < n - 1; i++)
    {
        min = i;
        for(j
= i + 1; j < n; j++)
        {
            if(s[j].per < s[min].per)
min = j;
        }
        temp =
s[i];    s[i] =
s[min];    s[min] =
temp;
    }
}

int main()
{
    FILE *fp, *out;
    struct Stud s[100];
    int n = 0, i;

    fp = fopen("Student.txt", "r");
    if(fp == NULL)
    {
```

```

        printf("File not found");
return 0;
    }

    while(n < 100 && fscanf(fp, "%s %d %f", s[n].name,
&s[n].age, &s[n].per) == 3) n++;
    fclose(fp);

    if(n == 0)
{
    printf("No records");
return 0;
}

    selectionSort(s, n);

    out = fopen("sortedstudent.txt", "w");
if(out == NULL)
{
    printf("Cannot create output file");
return 0;
}

    printf("Sorted data:\n");
for(i = 0; i < n; i++)
{
    fprintf(out, "%s %d %.2f\n", s[i].name, s[i].age, s[i].per);
printf("%s %d %.2f\n", s[i].name, s[i].age, s[i].per);
}
fclose(out);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (Primary keys are underlined)

Doctor (dcode,doctor_name, specialization, address, phone_no,fees)

Patient (pcode, Patient_name, symptoms)

There exists a one-to-many relationship between Doctor and Patient.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Doctor(      dcode
INT PRIMARY KEY,      doctor_name
VARCHAR(50),      specialization
VARCHAR(50),      address
VARCHAR(100),      phone_no
VARCHAR(15),      fees
NUMERIC(8,2)
);

CREATE TABLE Patient(      pcode
INT PRIMARY KEY,
Patient_name VARCHAR(50),
symptoms VARCHAR(100),
      dcode INT REFERENCES Doctor(dcode)
);

```

```

INSERT INTO Doctor VALUES
(1,'Dr. Patil','Neurologist','Sadashiv Peth','9822000001',500),
(2,'Dr. Shah','Orthopedic','Shivajinagar','9822000002',300),
(3,'Dr. Rao','General','Camp','9822000003',200);

```

```

INSERT INTO Patient VALUES
(1,'Meena','Fever',1),
(2,'Mohan','Fracture',2),
(3,'Anil','Fever',3);

```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Find the names of all Patients which start with “M”.

Answer:

```

SELECT Patient_name FROM Patient WHERE
Patient_name LIKE 'M%';

```

ii. Count the number of doctors who are Neurologists.

Answer:

```

SELECT COUNT(*) FROM Doctor
WHERE specialization = 'Neurologist';

```

iii. Give the list of all patients who are suffering from “Fever”

Answer:

```
SELECT * FROM Patient
WHERE symptoms = 'Fever';
```

iv. Find the specialization and phone numbers of all doctors from 'Sadashiv Peth'.

Q 2(C)

C. Write a query to display the list of all patients who are treated by doctors staying in 'Shivajinagar' area with the specialization "Orthopedic". [4 marks]

Answer:

```
SELECT P.*
FROM Patient P
JOIN Doctor D ON P.dcode = D.dcode
WHERE D.address = 'Shivajinagar'
AND D.specialization = 'Orthopedic';
```

Q 3) Viva [5 marks]